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Regel:

$$\begin{aligned}f(x) &= \frac{g(x)}{h(x)} \\ \Rightarrow f'(x) &= \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2} \\ f(x) &= \frac{x^3 - 1}{x^2 + x} \\ \Rightarrow f'(x) &= \frac{(3x^2)(x^2 + x) - (x^3 - 1)(2x + 1)}{(x^2 + x)^2} = \frac{(3x^4 + 3x^3) - (2x^4 + x^3 - 2x - 1)}{x^4 + 2x^3 + x^2} \\ &= \frac{x^4 + 2x^3 + 2x + 1}{x^4 + 2x^3 + x^2}\end{aligned}$$

References